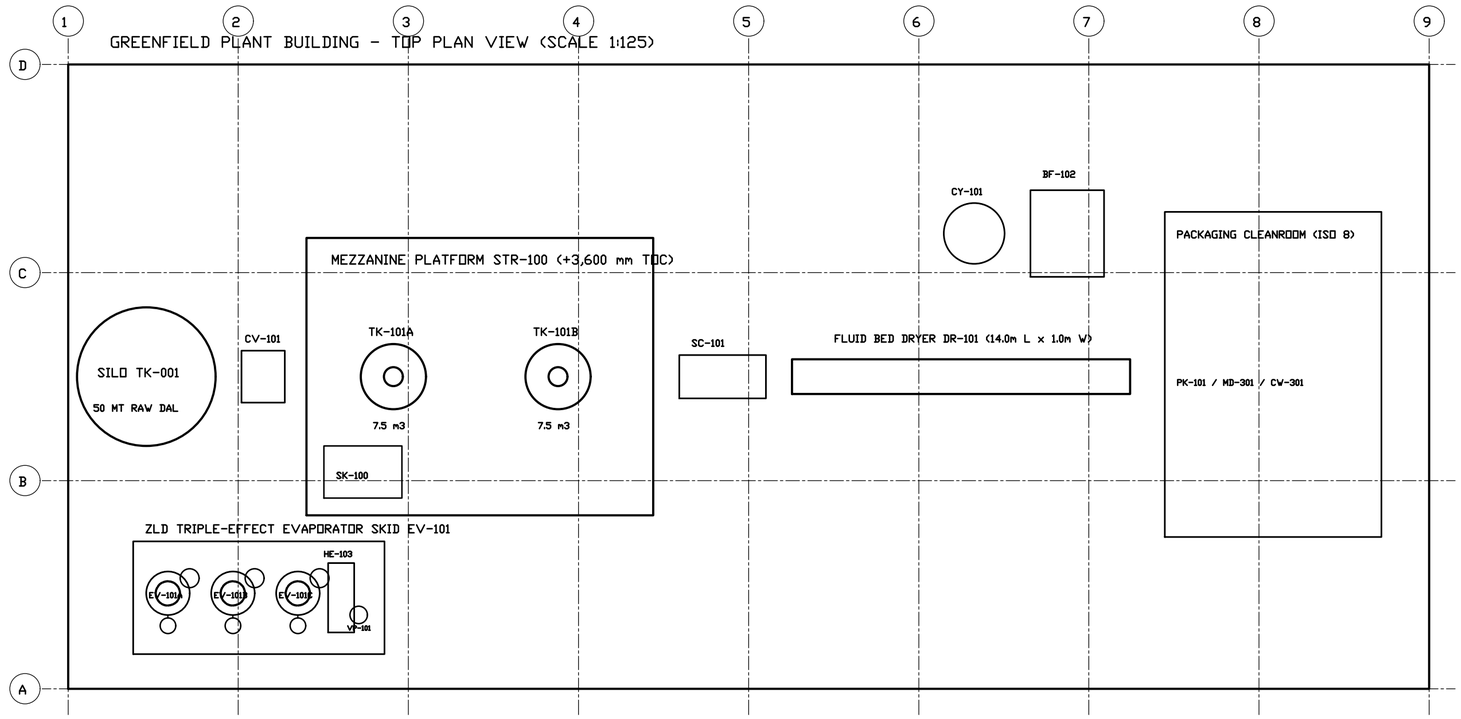
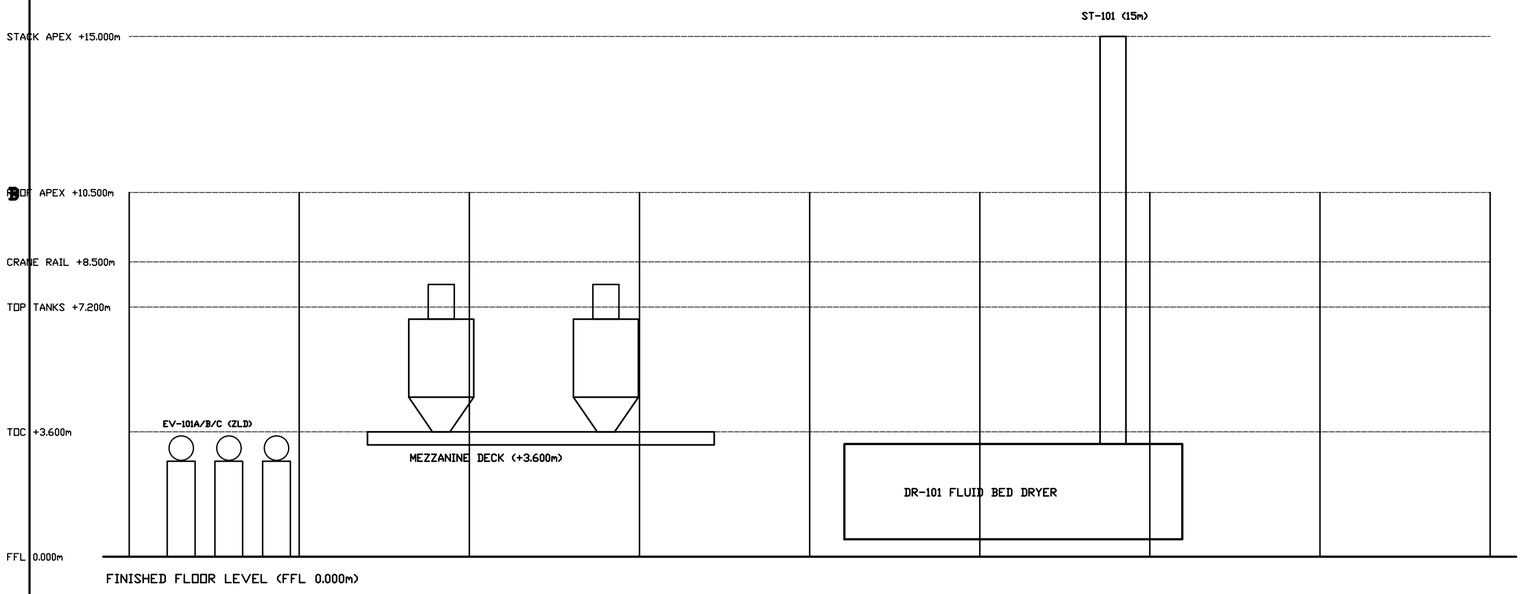




MASTER EQUIPMENT SCHEDULE & OPERATING WEIGHTS

TAG	EQUIPMENT DESCRIPTION	DIMENSIONS (mm)	ELEV.	OP. WT	POWER	MOC
TK-001	Raw Grain Silo (50 MT)	Dia 4,500 x 6,500	0.000m	58,500 kg	2.2 kW	Carbon Steel
CV-101	Bucket Elevator Infeed	1,000 x 1,200 x 9,000	0.000m	1,850 kg	3.7 kW	SS304 Contact
SC-100	Grain Cleaner & Destoner	2,200 x 1,400 x 1,600	0.000m	950 kg	2.2 kW	SS304 Casing
TK-101A	Soak Vessel A (7.5 m3)	Dia 2,000 x 3,450	+3.600m	9,850 kg	4.0 kW	AISI SS316L
TK-101B	Soak Vessel B (7.5 m3)	Dia 2,000 x 3,450	+3.600m	9,850 kg	4.0 kW	AISI SS316L
SK-100	Chemical Dosing Skid	2,400 x 1,400 x 1,900	+3.600m	1,450 kg	1.5 kW	HDPE / SS316L
SC-101	Dewatering Screen	3,000 x 1,200 x 1,200	+3.600m	1,250 kg	3.0 kW	AISI SS316L
DR-101	Fluid Bed Dryer	14,000 x 1,000 x 2,800	0.000m	6,850 kg	6.0 kW	AISI SS304
HE-101	Steam Air Heater (Dryer)	1,800 x 1,200 x 1,500	0.000m	1,200 kg	N/A	Al-Finned SS
BL-101	Primary Air Blower	1,600 x 1,400 x 1,400	0.000m	680 kg	18.5 kW	Carbon Steel
CY-101	Cyclone Separators (2x)	Dia 1,200 x 4,000	0.000m	1,700 kg	N/A	AISI SS304
BF-102	Dust Collector Baghouse	2,800 x 2,200 x 4,800	0.000m	2,450 kg	N/A	Carbon Steel
PK-101	Packaging Machine Line	4,500 x 2,000 x 3,200	0.000m	3,200 kg	7.5 kW	AISI SS304
EV-101	Triple Effect ZLD Evap	7,500 x 3,000 x 4,500	0.000m	5,500 kg	11.0 kW	AISI SS316L
TR-01	Substation Transformer	2,200 x 1,600 x 2,000	0.000m	2,800 kg	500 kVA	Oil Immersed
DG-01	Standby Generator Yard	3,500 x 1,500 x 2,200	0.000m	3,500 kg	200 kW	Acoustic Encl

CIVIL FOUNDATION LOADS & STRUCTURAL CRITERIA (IS 800 / IS 456)

FOUNDATION ZONE	DESIGN LOAD	CONCRETE SPEC	ANCHOR BOLTS	SETTLEMENT TOL.
Raw Silo Yard (TK-001)	220 kN/m2 Static	M30 RCC Raft	8x M36 Gr 8.8 (750mm)	< 10 mm Allowable
Process Mezzanine (STR-100)	180 kN/m2 Factored	M25 Pedestals	16x M30 Gr 8.8 (600mm)	< 12 mm Differential
Fluid Bed Dryer (DR-101)	150 kN/m2 Dynamic	M25 Isolated Base	12x M24 Dynamic Anchor	< 5 mm Resonant Tol.
ZLD Evaporator Skid (EV-101)	160 kN/m2 Static	M25 Acid-Resistant	10x M24 SS316 Anchors	< 8 mm Uniform
Machinery Floor (Ground FFL)	25.0 kN/m2 Live	M25 200mm Slab	Welded Mesh B500	< 15 mm Long Term

REGULATORY CODES, OSHA & EMISSION COMPLIANCE REGISTER

REGULATORY CODE	APPLICABLE SYSTEM	MANDATED ENGINEERING CRITERIA	VERIFICATION STATUS
OSHA 1910.29	Mezzanine & Ladders	1,000mm clear walkways, 1,070mm top rail, 42kN/m toe board	100% DESIGN CERTIFIED
NFPA 68 / 69	Dryer Hood & Baghouse	2x 1.2 m2 explosion relief burst panels, Pred < 0.20 bar g	100% DESIGN CERTIFIED
ASME Sec VIII Div 1	TK-101 & Evaporator	Full radiographed welds, hydrotested @ 1.5x design MAWP	100% DESIGN CERTIFIED
CPCB ZLD Norms	Effluent Treatment	0.00 m3/day discharge, 96.2% distillate recovery closed loop	STATE SPCB COMPLIANT
IS 3043 / CEA 2010	Electrical Substation	Dual independent copper earth pits, 25 kA / 1s withstand	INSPECTION PASSED

ENGINEERING GOVERNANCE, DIGITAL TWIN & SOLIDWORKS PROVENANCE

AUDIT & PROVENANCE CRITERIA	VERIFIED SPECIFICATION & SYSTEM GROUND TRUTH
Lead Process Engineer	Sourish Senapati (FinnoaQ Agro-Industrial Technologies Ltd.)
Academic Guidance	Dr. Archisman Ray (Indian Institute of Technology Kharagpur)
Project Reference	FNQ-PR-2026-001	15 TPD Continuous Quick-Cook Pulse Facility
CAD Provenance Engine	Native Dassault Systemes SOLIDWORKS 2025 SP1.2 Parametric Solids
Assembly Mates Status	10 Subsystems Parametrically Positioned (0.00 mm Clash Envelope)
Thermodynamic Closure	Exact 0.0000% Mass Balance via Aspen Plus V14 NRTL Property Package
Zero Liquid Discharge	96.20% Closed-Loop Distillate Recovery (0.00 m3/day Liquid Effluent)

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

PLANT GENERAL ARRANGEMENT MASTER LAYOUT
DISCIPLINE: GENERAL ARRANGEMENT | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-014	REV: V8.0	SHEET: 13 OF 13
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED – PRELIMINARY – NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ